Harmonic Oscillator

Quantum Physics, 4th Exercise Sheet

Exercises

Exercise 1: Mass-and-spring system and a quantum dot as quantum harmonic oscillators

a) For a mass-and-spring system with $m = 1\,\mathrm{kg}$ and $k = 1\,\mathrm{N/m}$, calculate the eigenfrequency ω, the energy spacing $\hbar\omega$ between adjacent quantum energy levels, and the oscillator length $\ell = \sqrt{\frac{\hbar}{m\omega}}$. ($\hbar = 1.05 \times 10^{-34}$ Js $= 0.66$ μeV $\cdot$ ns, 1 eV $= 1.6 \times 10^{-19}$ J)

b) Consider an electron confined in a one-dimensional harmonic potential with oscillator length $\ell = 100\,\mathrm{nm}$, i.e., the Hamiltonian reads as

$$H = \frac{p^2}{2m} + \frac{\hbar^2}{2m\ell^4}x^2. \tag{1}$$

i) Compute the energy gap $\hbar\omega$ and express it in electronvolts.($m_e = 9.1 \cdot 10^{-31}$ kg)

ii) Switch on now a static electric field. This results in the potential: $V_{\mathcal{E}} = e\mathcal{E}x$, and the full Hamiltonian reads as

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2x^2 + e\mathcal{E}x. \tag{2}$$

Complete the square for x and conclude that above Hamiltonian describes a Harmonic oscillator shifted in space and in energy. Calculate the value of the electric field for which the center of the harmonic oscillator is shifted by 100 nm.

iii) Compute the force ($F_{\mathcal{E}} = e\mathcal{E}$) and electric field required to shift the ground-state energy by 1 meV.

Exercise 2: Ladder operators and number operator

The ladder operators are defined as

$$a = \frac{1}{\sqrt{2}\ell}(x + \tfrac{i}{m\omega}p), \quad a^\dagger = \frac{1}{\sqrt{2}\ell}(x - \tfrac{i}{m\omega}p).$$

a) Express the x and p operators with the ladder operators.

b) Use the canonical commutation relation $[x, p] = i\hbar I$ and the linearity of the commutation relation, to calculate the commutation relation $[a, a^\dagger]$.

c) Show that $H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2x^2 = \frac{\hbar\omega}{2}(a^\dagger a + aa^\dagger) = \hbar\omega\big(a^\dagger a + \frac{1}{2}I\big)$. *Hint: for the second equation use that* $AB = BA + [A, B]$.

d) Express the matrix representation of the operators a, $a^\dagger$, and the number operator $N = a^\dagger a$ in the truncated basis $\{|0\rangle, |1\rangle, |2\rangle\}$. I.e. calculate all possible matrix elements and order them into a matrix. *Hints:* $a|n\rangle = \sqrt{n}\,|n-1\rangle$, $a^\dagger|n\rangle = \sqrt{n+1}\,|n+1\rangle$.

e) Denote the above 3×3 matrices as $\tilde{a}$, $\tilde{a}^\dagger$, and $\tilde{N}$. Is $\tilde{a}^\dagger\tilde{a}$ equal to $\tilde{N}$?

f) Do $\tilde{a}$ and $\tilde{a}^\dagger$ satisfy the canonical commutator relation $[\tilde{a}, \tilde{a}^\dagger] = I$?

Exercise 3: Expectation values and Heisenberg uncertainty

a) Compute $\langle x\rangle$, $\langle p\rangle$, $\langle x^2\rangle$, and $\langle p^2\rangle$ in the ground state of the harmonic oscillator. Show that the Heisenberg uncertainty relation is fulfilled.

b) Consider the state $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. Compute the expectation values of position and momentum in this state.

c) Repeat for $|\chi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i\,|1\rangle)$.

Exercise 4: Time evolution and Ehrenfest theorem

a) For the state $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, compute its time evolution under the harmonic oscillator Hamiltonian.

b) From the time-dependent state, determine the time evolution of $\langle x\rangle(t) = \langle x\rangle_{\psi(t)}$ and $\langle p\rangle(t) = \langle p\rangle_{\psi(t)}$.

c) Check whether the time evolution satisfies the relation

$$m\,\frac{d\langle x\rangle_{\psi(t)}}{dt} = \langle p\rangle_{\psi(t)}.$$

d) Assume now, that the system is in a slightly shifted ground state: $\psi_0(x-\delta x)$. In the limit $\delta x \ll \ell$, we can approximate the state up to first order using Taylor-series: $\psi_0(x-\delta x) \approx \psi_0 - \delta x \partial_x \psi_0$. Using the form of the momentum operator: $p = -i\hbar\partial_x$ and the ladder operators, express the state $\psi_0(x-\delta x)$ in the eigenbasis of the harmonic oscillator.

e) Calculate the time-evolution of the state, and calculate the time evolution of the expectation value of x.

Exercise 5: First excited state wave function (homework)

a) Using (i) the ground-state wave function

$$\psi_0(x) = \frac{1}{\pi^{1/4}\sqrt{\ell}}\, e^{-x^2/2\ell^2},$$

(ii) the relation $a^\dagger|0\rangle = |1\rangle$, and (iii) the operator form of $a^\dagger$, derive the explicit form of $\psi_1(x)$.

Exercise 6: Coherent states (homework)

a) Recall that coherent states are defined by $a|\alpha\rangle = \alpha|\alpha\rangle$ with $\alpha \in \mathbb{C}$.

b) Compute the expectation values $\langle x\rangle$ and $\langle p\rangle$ in a coherent state $|\alpha\rangle$.

c) Compute the standard deviations Δx and Δp for $|\alpha\rangle$.

d) Determine the time evolution of a coherent state $|\alpha_0\rangle$ under the harmonic oscillator Hamiltonian and show that

$$|\psi(t)\rangle = e^{-i\omega t/2}\left|e^{-i\omega t}\alpha_0\right\rangle.$$

e) Write down the time evolution of $\langle x\rangle$ and $\langle p\rangle$ and compare it to classical oscillatory motion.

f) Discuss whether the raising operator $a^\dagger$ has any eigenvectors.